# Gabriel Jacob Perin

gabrieljp@usp.br    gabjp.github.io    gabjp    Google Scholar

## Research Interests

I am broadly interested in **Large Language Models (LLMs)**, **Geometric Deep Learning (GDL)**, and **AI for Science (AI4Science)**. My research interests include efficient , interpretable, safe, and robust machine learning methods, with applications in scientific domains such as astronomy, materials science, and healthcare.

## Education

**MS** **University of São Paulo**, Computer Science — March 2026–December 2027 (Expected)

- Advisors: Prof. Nina S. T. Hirata (IME-USP), Dr. André Araujo (Google DeepMind).

**BS** **University of São Paulo**, Computer Science — March 2021–December 2025

- GPA: 9.2/10
- **Coursework:** Machine Learning, Optimization, Algorithms, Discrete Mathematics.
- **Thesis**: An Introduction to Geometric Deep Learning on Sets, Graphs, and Grids.

## Research Experience

**University of São Paulo**, Graduate Research Assistant — SP, BR, March 2026–Present

- Advisors: Prof. Nina S. T. Hirata and Dr. André Araujo.
- Researching efficient and training-free methods for controlling LLMs, with applications to post-training interventions and model behavior editing.

**IBM Research**, Research Intern — SP, BR, November 2024–November 2025

- Manager: Dr. Mathias Steiner.
- Worked on machine-learning interatomic potentials (MLIPs) and foundation models for materials science.

**University of Texas at Austin**, Research Intern — TX, USA, April–July 2024

- Advisor: Prof. Zhangyang “Atlas” Wang.
- Researched model merging methods for Large Language Models (LLMs).
- Investigated the robustness of safety alignment under benign and malicious fine-tuning.
- Collaborated with the group remotely (May 2023–April 2025) in addition to the official research visit (April–July 2024).

**University of São Paulo**, Undergraduate Research Assistant — SP, BR, February 2022–April 2025

- Advisor: Prof. Nina S. T. Hirata.
- Developed machine learning models (Random Forests, CNNs, and self-supervised methods) for classifying astronomical objects in the S-PLUS survey, co-advised by Prof. Claudia L. M. de Oliveira.
- Designed few-shot retinal disease classification pipelines using meta-learning algorithms (Reptile) with CNN and Vision Transformer backbones.

## Teaching Experience

**University of São Paulo**, Teaching Assistant — SP, BR
March 2025–June 2025

- Advisor: Prof. Nina S. T. Hirata.
- Course: Introduction to Machine Learning (undergraduate).
- Held office hours and provided detailed feedback on assignments.

## Publications and Preprints

**Training-Free Task Vectors for LLM Behavioral Control** — 2026
**G. Jacob Perin**, L. Boscaini, A. Araujo, N. S. T. Hirata
In Submission

**Equivariant Interatomic Potentials without Tensor Products** — 2026
T. Reschützegger, S. Aykent, **G. Jacob Perin**, B. H. Nunes, F. Cipcigan, R. N. B. Ferreira, M. Steiner, F. L. Thiemann
Preprint

**Meta-Learning based Few-Shot Classification of Retinal Diseases** — 2026
**G. Jacob Perin**, D. J. C. Alves, L. M. S. Sousa, E. M. de Elias, N. S. T. Hirata
Accepted on Journal of the Brazilian Computer Society (JBCS)

**LAD-VF: LLM-Automatic Differentiation Enables Fine-Tuning-Free Robot Planning from Formal Methods Feedback** — 2026
Y. Yang, J. Hong, **G. Jacob Perin**, Z. Fan, L. Yin, Z. Wang, U. Topcu
ICRA 2026

**LoX: Low-Rank Extrapolation Robustifies LLM Safety Against Fine-tuning** — 2025
**G. Jacob Perin**, R. Chen, X. Chen, N. S. T. Hirata, Z. Wang, J. Hong
COLM 2025

**Extracting and understanding the superficial knowledge in alignment** — 2025
R. Chen, **Gabriel Jacob Perin**, X. Chen, X. Chen, Y. Han, N. S. T. Hirata, J. Hong, B. Kailkhura
NAACL 2025

**Few-shot Retinal Disease Classification on the Brazilian Multilabel Ophthalmological Dataset** — 2024
**G. Jacob Perin**, Nina S. T. Hirata
SIBGRAPI 2024

**RankMean: Module-Level Importance Score for Merging Fine-tuned LLM Models** — 2024
**G. Jacob Perin**, X. Chen, S. Liu, B. Kailkhura, Z. Wang, B. Gallagher
ACL 2024 Findings (short paper)

**The Fourth S-PLUS Data Release: 12-filter photometry covering 3000 square degrees in the Southern Hemisphere** — 2024
F. R. Herpich, ..., **G. Jacob Perin**, et al.
Astronomy and Astrophysics (A&A)

**Combinação de Dados Tabulares e Imagens para a Classificação de Objetos Astronômicos** — 2023
**G. Jacob Perin**, L. Nakazono, C. Mendes de Oliveira, N. S. T. Hirata
SIBGRAPI 2023, Workshop of Undergraduate Works (WUW)

## Awards and Grants

Honorable Mention, IME-USP Graduation Ceremony, University of São Paulo, 2026. Awarded for outstanding academic performance among the graduating class.

FAPESP Master's Fellowship, April 2026–February 2028

CAPES Master's Fellowship, March 2026

FAPESP International Fellowship for Research Experience for Undergraduates (BEPE), April–July 2024

Highlight of the Intermediate Phase, International Symposium of Scientific and Technological Initiation of the University of São Paulo (SIICUSP), 2023

FAPESP Scholarship for Research Experience for Undergraduates, October 2022–November 2024

## Event Presentations

**LAD-VF: LLM-Automatic Differentiation Enables Fine-Tuning-Free Robot Planning from Formal Methods Feedback** 2026

Y. Yang, J. Hong, **G. Jacob Perin**, Z. Fan, L. Yin, Z. Wang, U. Topcu

Poster, ICRA 2026

**LoX: Low-Rank Extrapolation Robustifies LLM Safety Against Fine-tuning** 2025

**G. Jacob Perin**, R. Chen, X. Chen, N. S. T. Hirata, Z. Wang, J. Hong

Poster, COLM 2025

**Few-shot Retinal Disease Classification on the Brazilian Multilabel Ophthalmological Dataset** 2024

**G. Jacob Perin**, N. S. T. Hirata

Oral, SIBGRAPI 2024

**RankMean: Module-Level Importance Score for Merging Fine-tuned LLM Models** 2024

**G. Jacob Perin**, X. Chen, S. Liu, B. Kailkhura, Z. Wang, B. Gallagher

Poster, ACL 2024 (digital)

**Combinação de Dados Tabulares e Imagens para a Classificação de Objetos Astronômicos** 2023

**G. Jacob Perin**, L. Nakazono, C. Mendes de Oliveira, N. S. T. Hirata

Poster, SIBGRAPI/WUW 2023 & SIICUSP 2023

## Others

**Languages:** Portuguese (native), English (fluent), Spanish (basic)

Founding member, L.E.A.R.N., machine learning group advised by Prof. Nina S. T. Hirata

**Technical Skills:** Python (PyTorch, PyTorch Lightning, TensorFlow, Pandas, Scikit-learn), C++, Git, Bash, SQL